GEPA Metric Optimization Results

Model: Qwen3-4B | Dataset: gsm8k | Samples: 16 | Acc: 0.0680

Model: Qwen3-4B | Dataset: gsm8k | Samples: 32 | Acc: 0.0720

Model: Qwen3-4B | Dataset: gsm8k | Samples: 64 | Acc: 0.0560

Model: Qwen3-4B | Dataset: gsm8k | Samples: 128 | Acc: 0.1040

Model: Qwen3-4B | Dataset: gsm8k | Samples: 256 | Acc: 0.0840

Model: Qwen3-4B | Dataset: superglue | Samples: 16 | Acc: 0.0000

Model: Qwen3-4B | Dataset: superglue | Samples: 32 | Acc: 0.0000

Model: Qwen3-4B | Dataset: superglue | Samples: 64 | Acc: 0.0357

Model: Qwen3-4B | Dataset: superglue | Samples: 128 | Acc: 0.0000

Model: Qwen3-4B | Dataset: superglue | Samples: 256 | Acc: 0.1964

Model: Qwen3-4B | Dataset: financial_phrasebank | Samples: 16 | Acc: 0.6750

Model: Qwen3-4B | Dataset: financial_phrasebank | Samples: 32 | Acc: 0.2550

Model: Qwen3-4B | Dataset: financial_phrasebank | Samples: 64 | Acc: 0.2500

Model: Qwen3-4B | Dataset: financial_phrasebank | Samples: 128 | Acc: 0.0700

Model: Qwen3-4B | Dataset: financial_phrasebank | Samples: 256 | Acc: 0.2500

Model: Llama-3-8B | Dataset: gsm8k | Samples: 16 | Acc: 0.0320

Model: Llama-3-8B | Dataset: gsm8k | Samples: 32 | Acc: 0.0600

Model: Llama-3-8B | Dataset: gsm8k | Samples: 64 | Acc: 0.0680

Model: Llama-3-8B | Dataset: gsm8k | Samples: 128 | Acc: 0.0520

Model: Llama-3-8B | Dataset: gsm8k | Samples: 256 | Acc: 0.0560

Model: Llama-3-8B | Dataset: superglue | Samples: 16 | Acc: 0.0000

Model: Llama-3-8B | Dataset: superglue | Samples: 32 | Acc: 0.0000

Model: Llama-3-8B | Dataset: superglue | Samples: 64 | Acc: 0.0000

Model: Llama-3-8B | Dataset: superglue | Samples: 128 | Acc: 0.0000

Model: Llama-3-8B | Dataset: superglue | Samples: 256 | Acc: 0.0179

Model: Llama-3-8B | Dataset: financial_phrasebank | Samples: 16 | Acc: 0.2150

Model: Llama-3-8B | Dataset: financial_phrasebank | Samples: 32 | Acc: 0.2050

Model: Llama-3-8B | Dataset: financial_phrasebank | Samples: 64 | Acc: 0.2150

Model: Llama-3-8B | Dataset: financial_phrasebank | Samples: 128 | Acc: 0.2350

Model: Llama-3-8B | Dataset: financial_phrasebank | Samples: 256 | Acc: 0.1950

Model: Llama-3-8B-Instruct | Dataset: gsm8k | Samples: 16 | Acc: 0.0640

Model: Llama-3-8B-Instruct | Dataset: gsm8k | Samples: 32 | Acc: 0.0200

Model: Llama-3-8B-Instruct | Dataset: gsm8k | Samples: 64 | Acc: 0.0040

Model: Llama-3-8B-Instruct | Dataset: gsm8k | Samples: 128 | Acc: 0.0920

Model: Llama-3-8B-Instruct | Dataset: gsm8k | Samples: 256 | Acc: 0.0600

Model: Llama-3-8B-Instruct | Dataset: superglue | Samples: 16 | Acc: 0.1786

Model: Llama-3-8B-Instruct | Dataset: superglue | Samples: 32 | Acc: 0.3393

Model: Llama-3-8B-Instruct | Dataset: superglue | Samples: 64 | Acc: 0.5357

Model: Llama-3-8B-Instruct | Dataset: superglue | Samples: 128 | Acc: 0.0000

Model: Llama-3-8B-Instruct | Dataset: superglue | Samples: 256 | Acc: 0.0179

Model: Llama-3-8B-Instruct | Dataset: financial_phrasebank | Samples: 16 | Acc: 0.2000

Model: Llama-3-8B-Instruct | Dataset: financial_phrasebank | Samples: 32 | Acc: 0.7400

Model: Llama-3-8B-Instruct | Dataset: financial_phrasebank | Samples: 64 | Acc: 0.2850

Model: Llama-3-8B-Instruct | Dataset: financial_phrasebank | Samples: 128 | Acc: 0.8150

Model: Llama-3-8B-Instruct | Dataset: financial_phrasebank | Samples: 256 | Acc: 0.7650

Model: Qwen3-8B | Dataset: gsm8k | Samples: 16 | Acc: 0.0960

Model: Qwen3-8B | Dataset: gsm8k | Samples: 32 | Acc: 0.1480

Model: Qwen3-8B | Dataset: gsm8k | Samples: 64 | Acc: 0.1600

Model: Qwen3-8B | Dataset: gsm8k | Samples: 128 | Acc: 0.1600

Model: Qwen3-8B | Dataset: gsm8k | Samples: 256 | Acc: 0.0400

Model: Qwen3-8B | Dataset: superglue | Samples: 16 | Acc: 0.5000

Model: Qwen3-8B | Dataset: superglue | Samples: 32 | Acc: 0.5000

Model: Qwen3-8B | Dataset: superglue | Samples: 64 | Acc: 0.0000

Model: Qwen3-8B | Dataset: superglue | Samples: 128 | Acc: 0.0000

Model: Qwen3-8B | Dataset: superglue | Samples: 256 | Acc: 0.0000

Model: Qwen3-8B | Dataset: financial_phrasebank | Samples: 16 | Acc: 0.3500

Model: Qwen3-8B | Dataset: financial_phrasebank | Samples: 32 | Acc: 0.0250

Model: Qwen3-8B | Dataset: financial_phrasebank | Samples: 64 | Acc: 0.2800

Model: Qwen3-8B | Dataset: financial_phrasebank | Samples: 128 | Acc: 0.2050

Model: Qwen3-8B | Dataset: financial_phrasebank | Samples: 256 | Acc: 0.2300

Model: Gemma-2-9B | Dataset: gsm8k | Samples: 16 | Acc: 0.0240

Model: Gemma-2-9B | Dataset: gsm8k | Samples: 32 | Acc: 0.0240

Model: Gemma-2-9B | Dataset: gsm8k | Samples: 64 | Acc: 0.0240

Model: Gemma-2-9B | Dataset: gsm8k | Samples: 128 | Acc: 0.0240

Model: Gemma-2-9B | Dataset: gsm8k | Samples: 256 | Acc: 0.0240

Model: Gemma-2-9B | Dataset: superglue | Samples: 16 | Acc: 0.0000

Model: Gemma-2-9B | Dataset: superglue | Samples: 32 | Acc: 0.0000

Model: Gemma-2-9B | Dataset: superglue | Samples: 64 | Acc: 0.0000

Model: Gemma-2-9B | Dataset: superglue | Samples: 128 | Acc: 0.0000

Model: Gemma-2-9B | Dataset: superglue | Samples: 256 | Acc: 0.0000

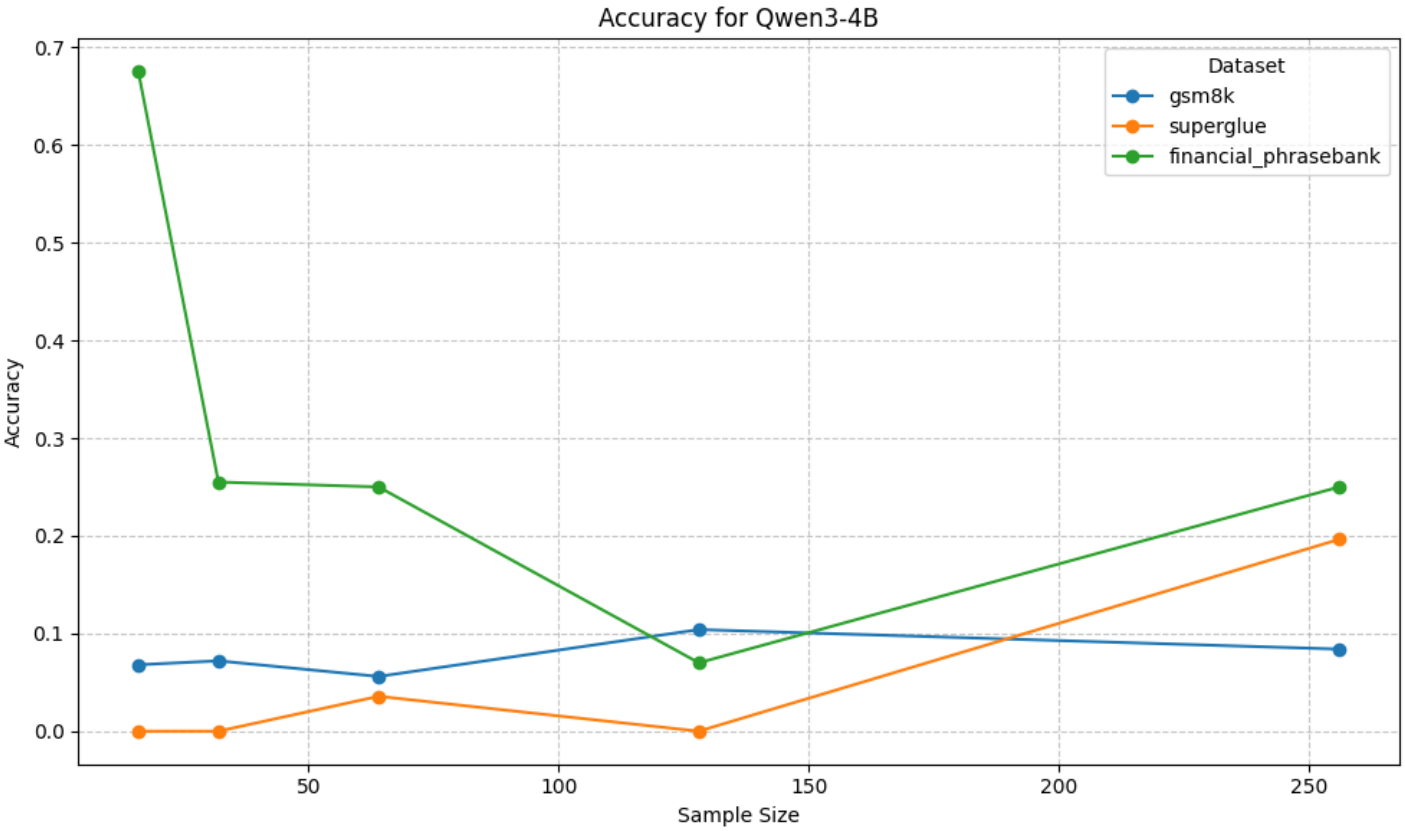
Model: Gemma-2-9B | Dataset: financial_phrasebank | Samples: 16 | Acc: 0.0900

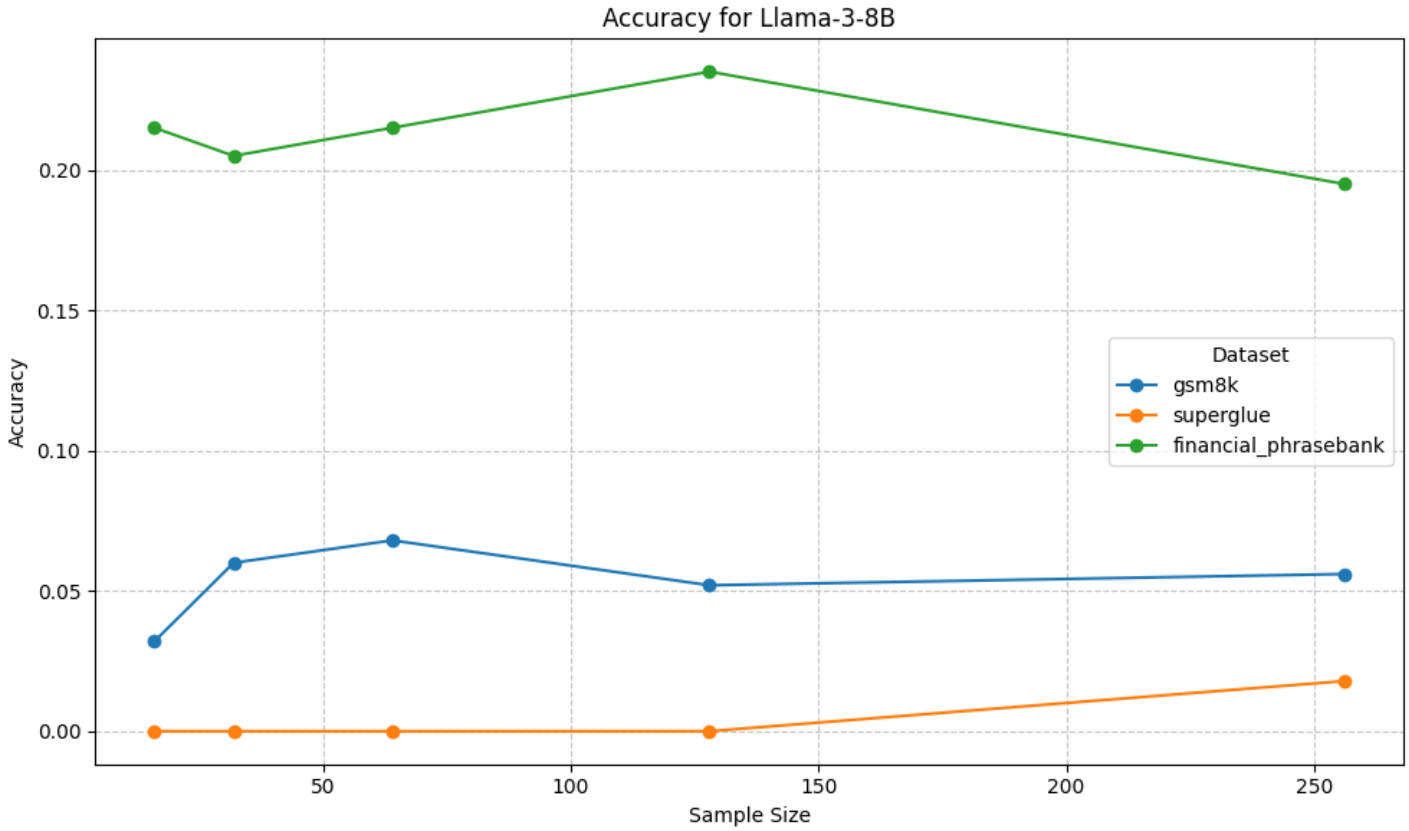
Model: Gemma-2-9B | Dataset: financial_phrasebank | Samples: 32 | Acc: 0.0900

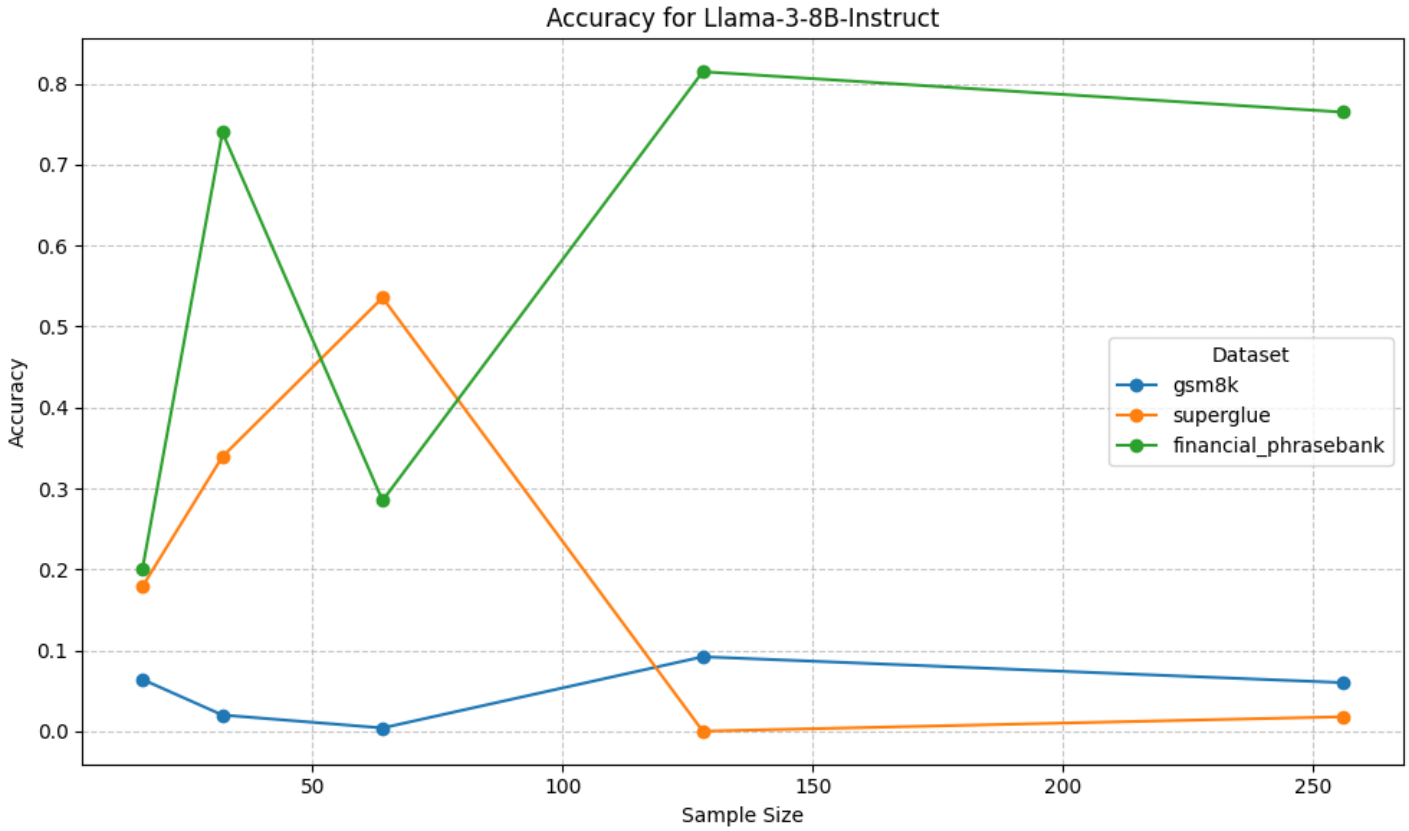
Model: Gemma-2-9B | Dataset: financial_phrasebank | Samples: 64 | Acc: 0.0900

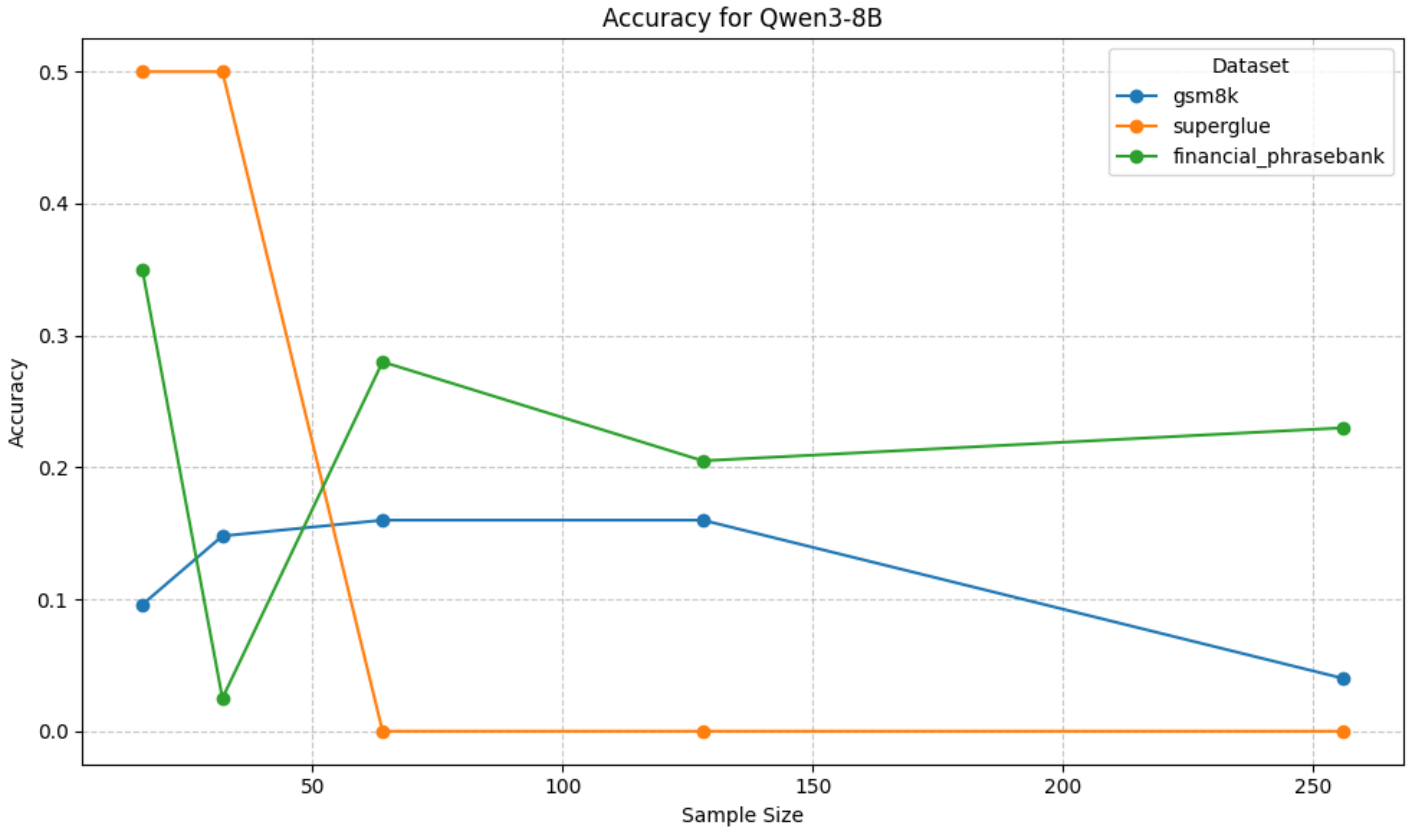
Model: Gemma-2-9B | Dataset: financial_phrasebank | Samples: 128 | Acc: 0.0900

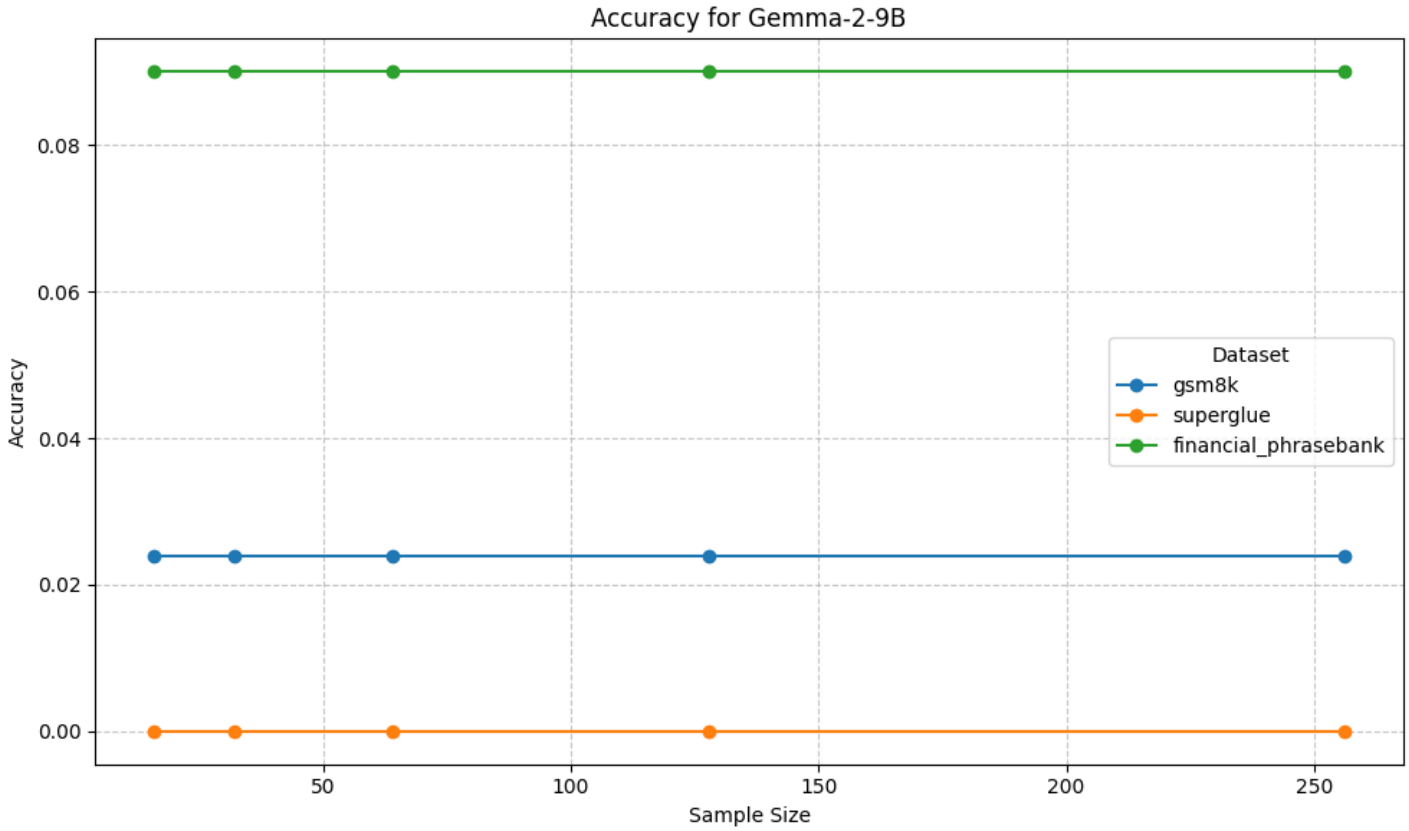
Model: Gemma-2-9B | Dataset: financial_phrasebank | Samples: 256 | Acc: 0.0900

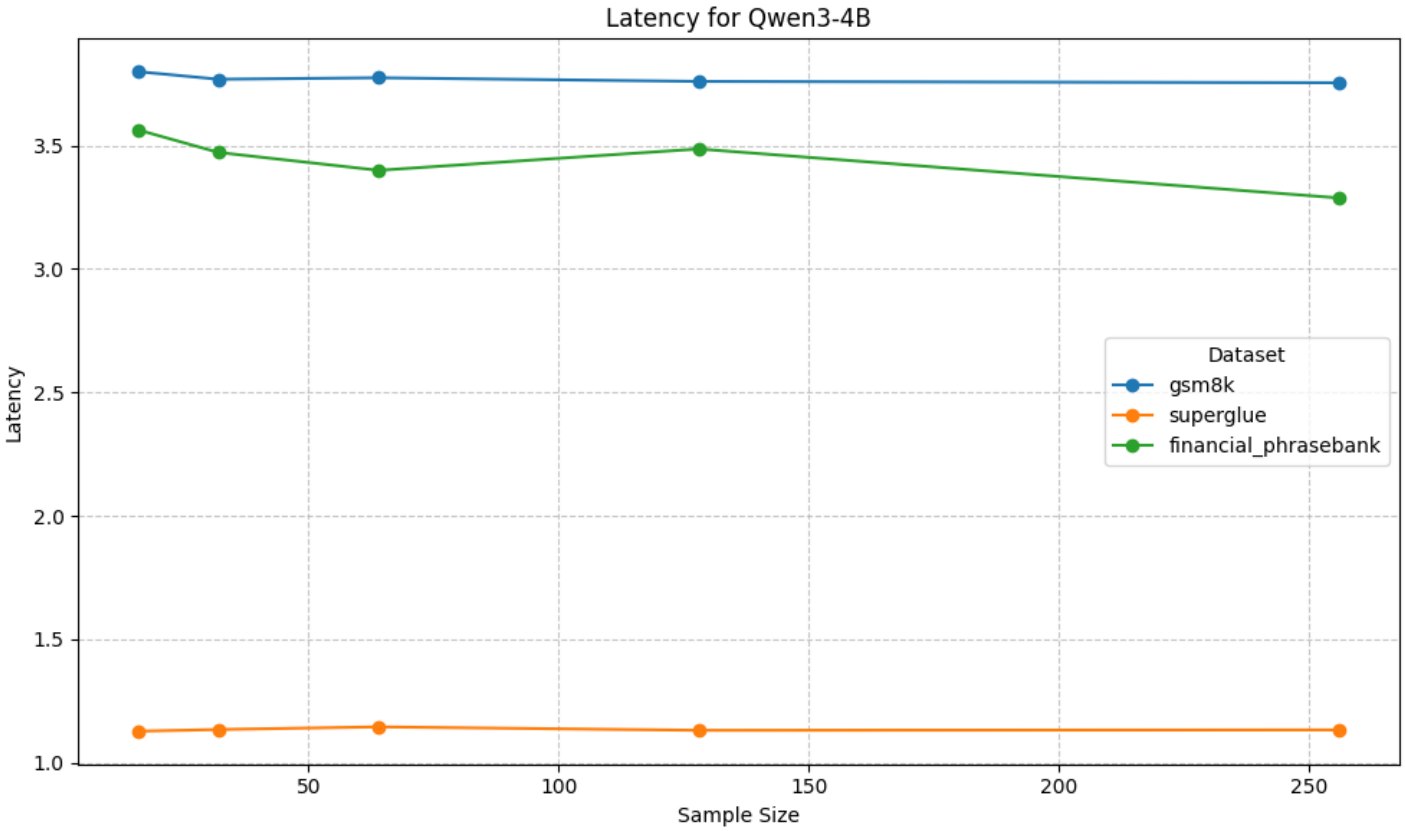

Accuracy for Qwen3-4B
Dataset
gsm8k
superglue
financial_phrasebank
Accuracy
Sample Size
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
50
100
150
200
250

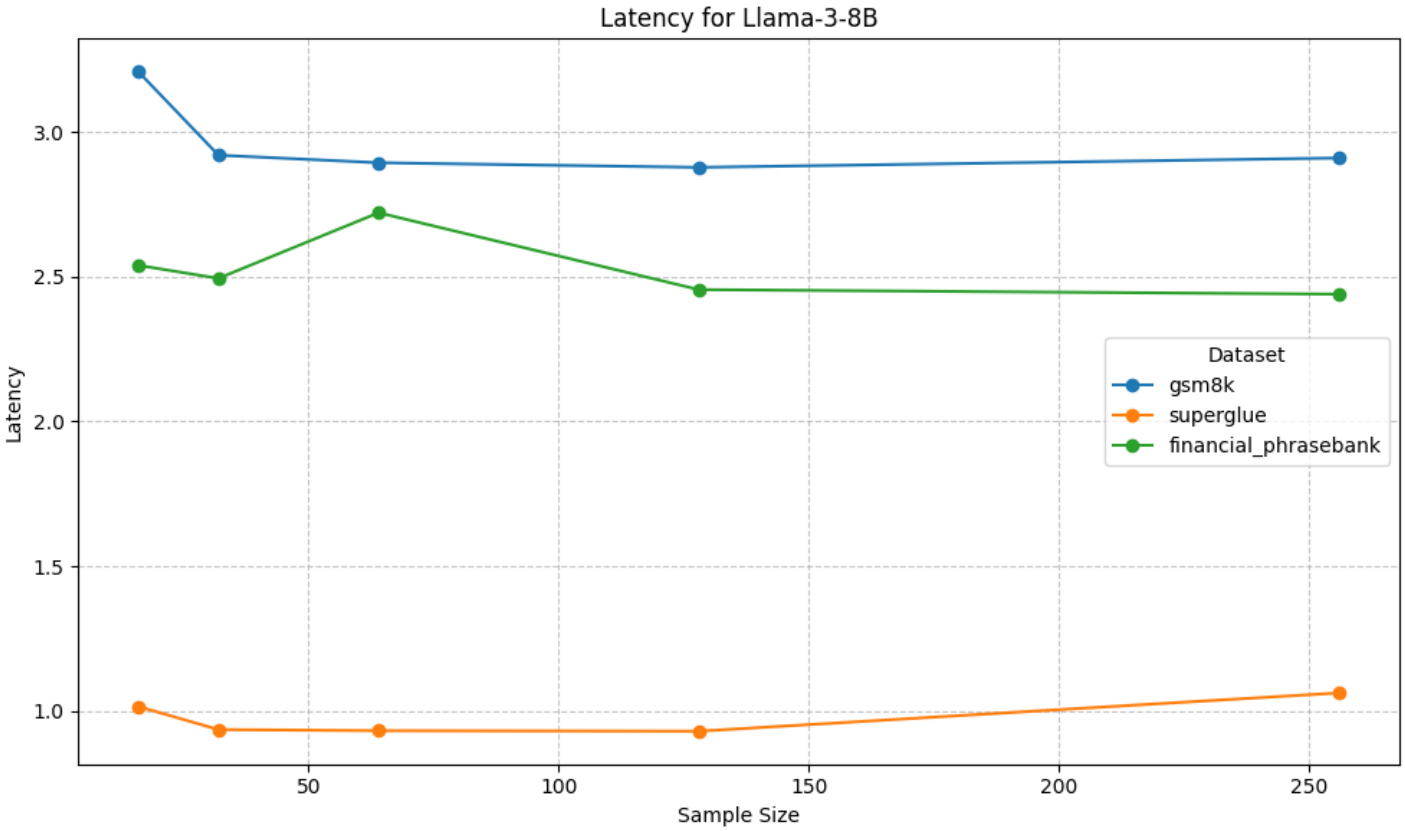

Accuracy for Llama-3-8B
Accuracy
Sample Size
0.00
0.05
0.10
0.15
0.20
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

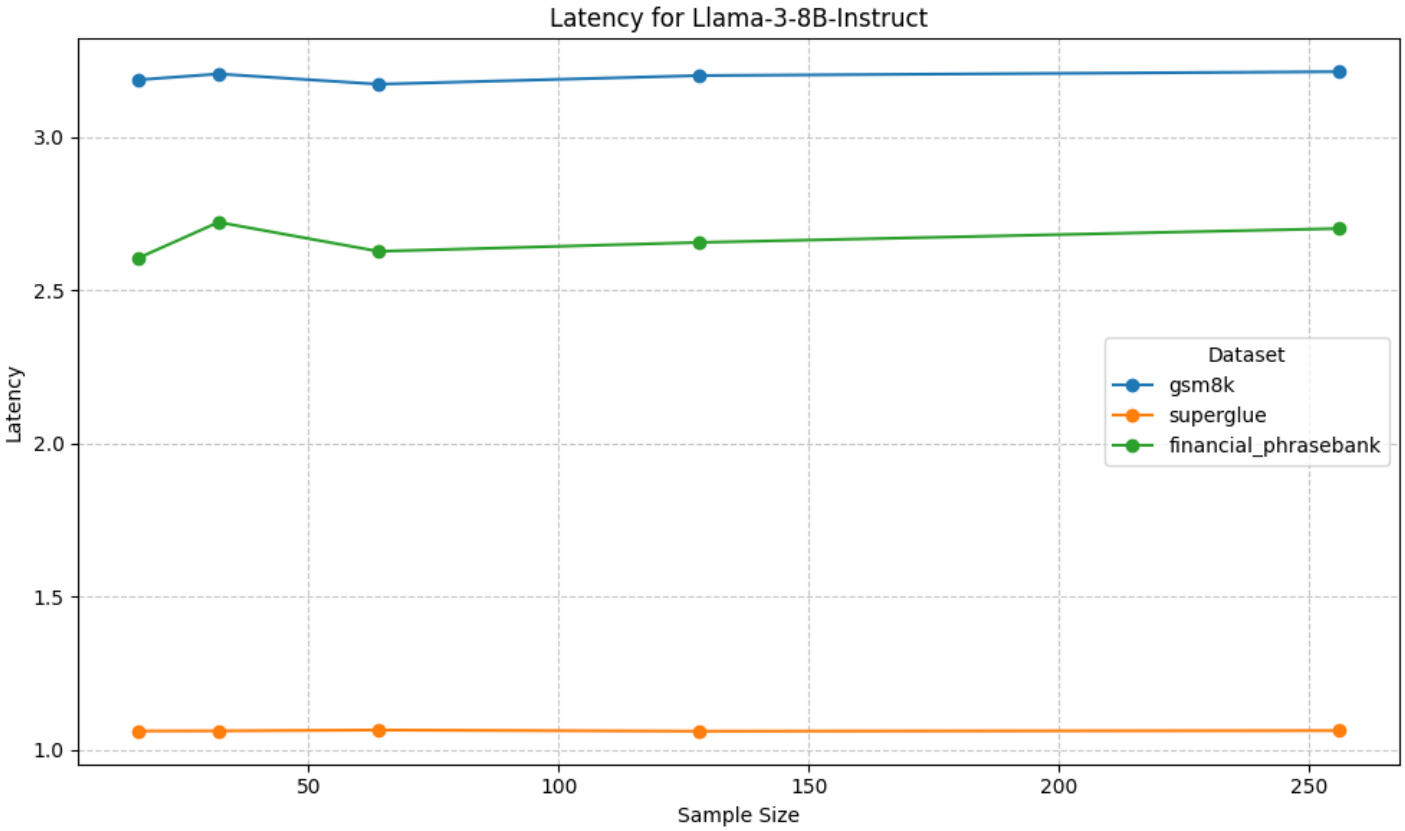

Accuracy for Llama-3-8B-Instruct
Accuracy
Sample Size
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

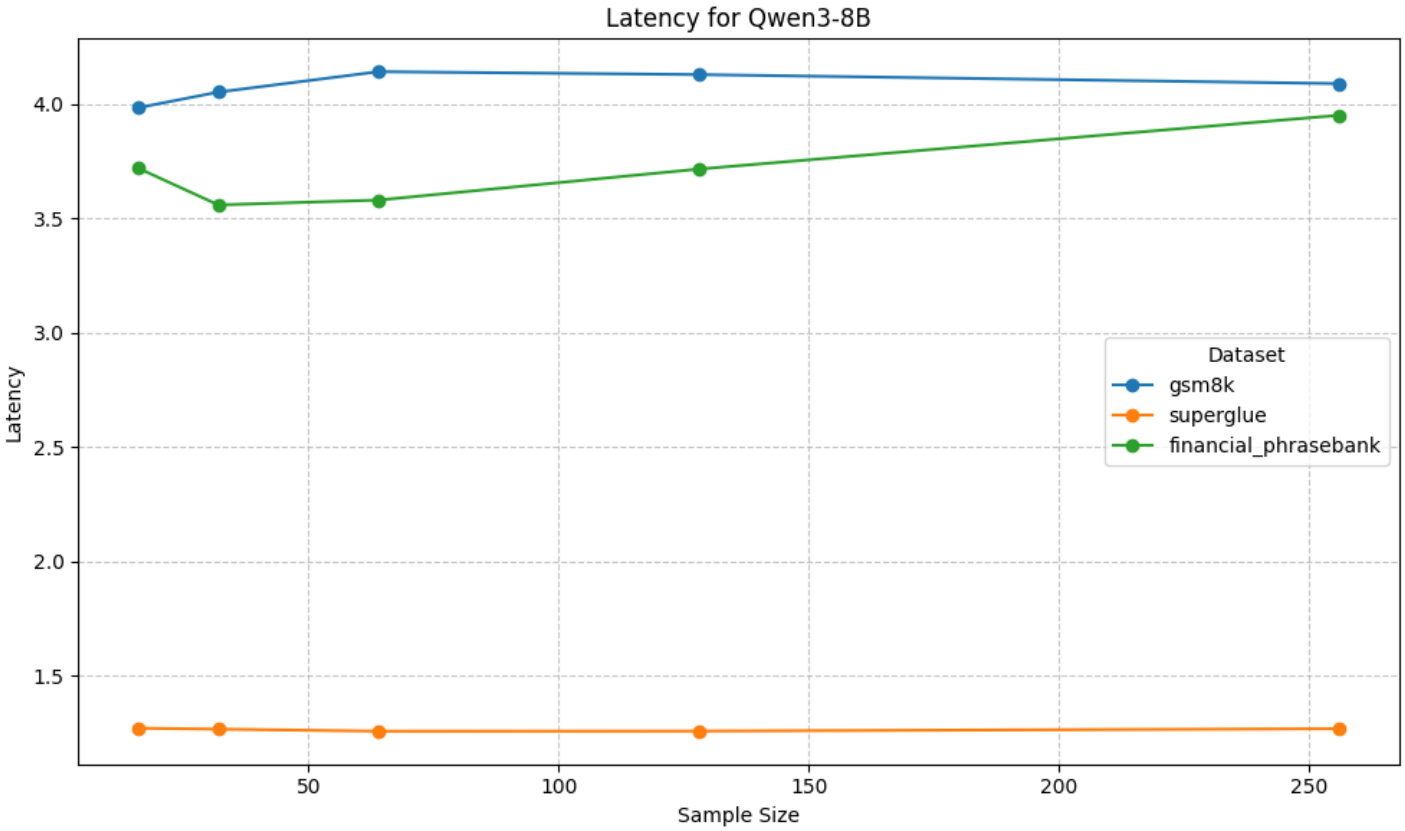

Accuracy for Qwen3-8B
Dataset
gsm8k
superglue
financial_phrasebank
Accuracy
Sample Size
0.0
0.1
0.2
0.3
0.4
0.5
50
100
150
200
250

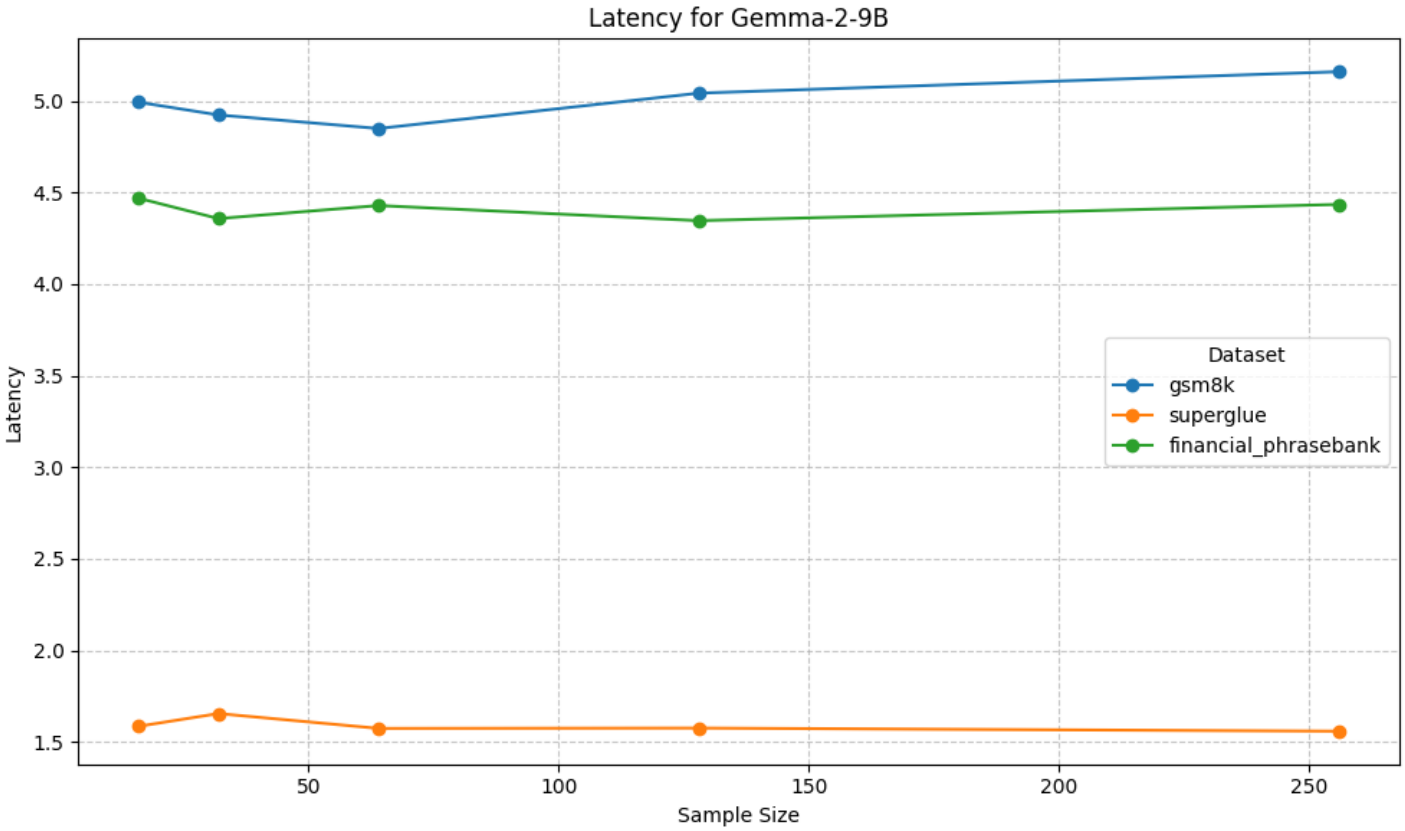

Accuracy for Gemma-2-9B
Accuracy
0.00
0.02
0.04
0.06
0.08
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank

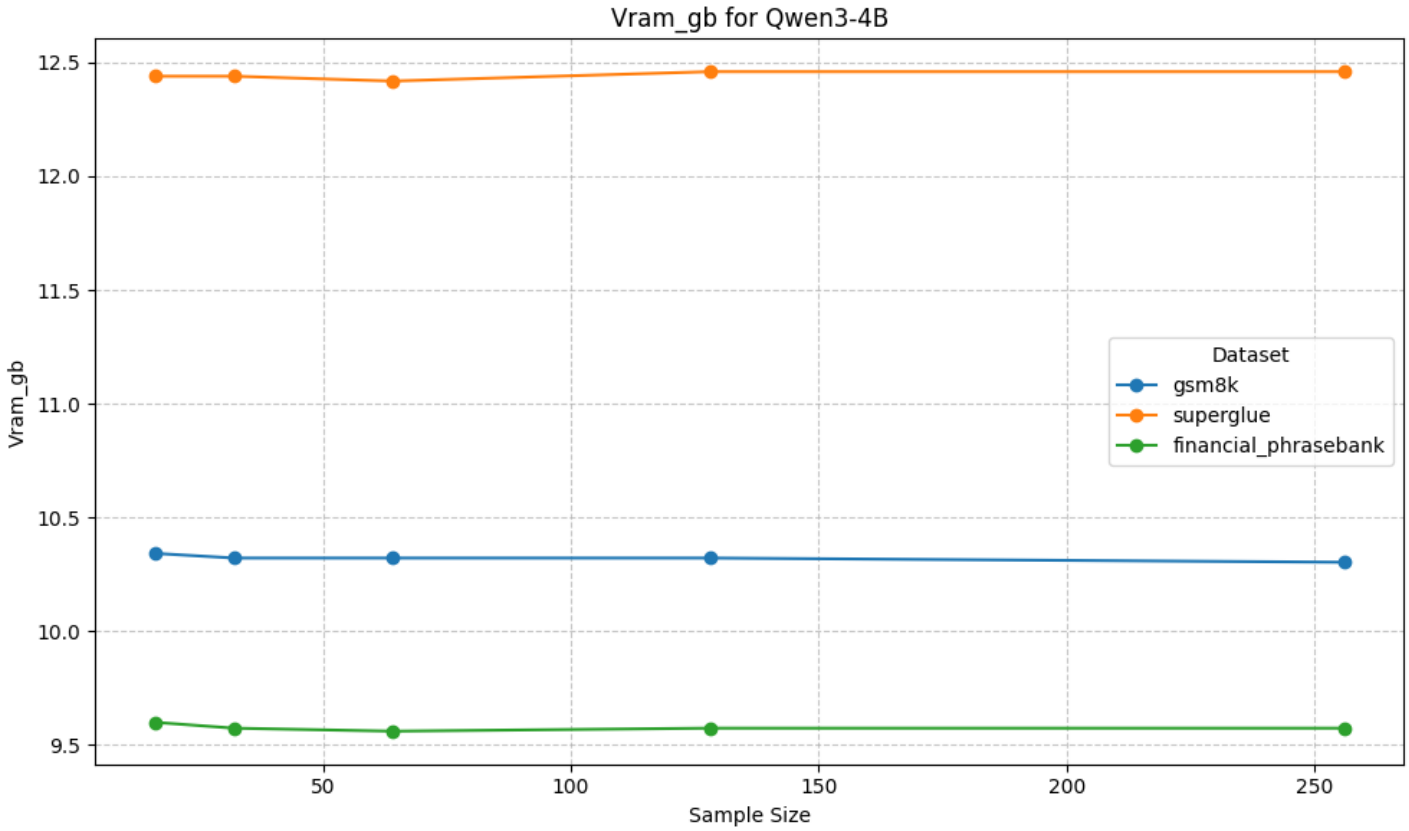

Latency for Qwen3-4B
Latency
Sample Size
3.5
3.0
2.5
2.0
1.5
1.0
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

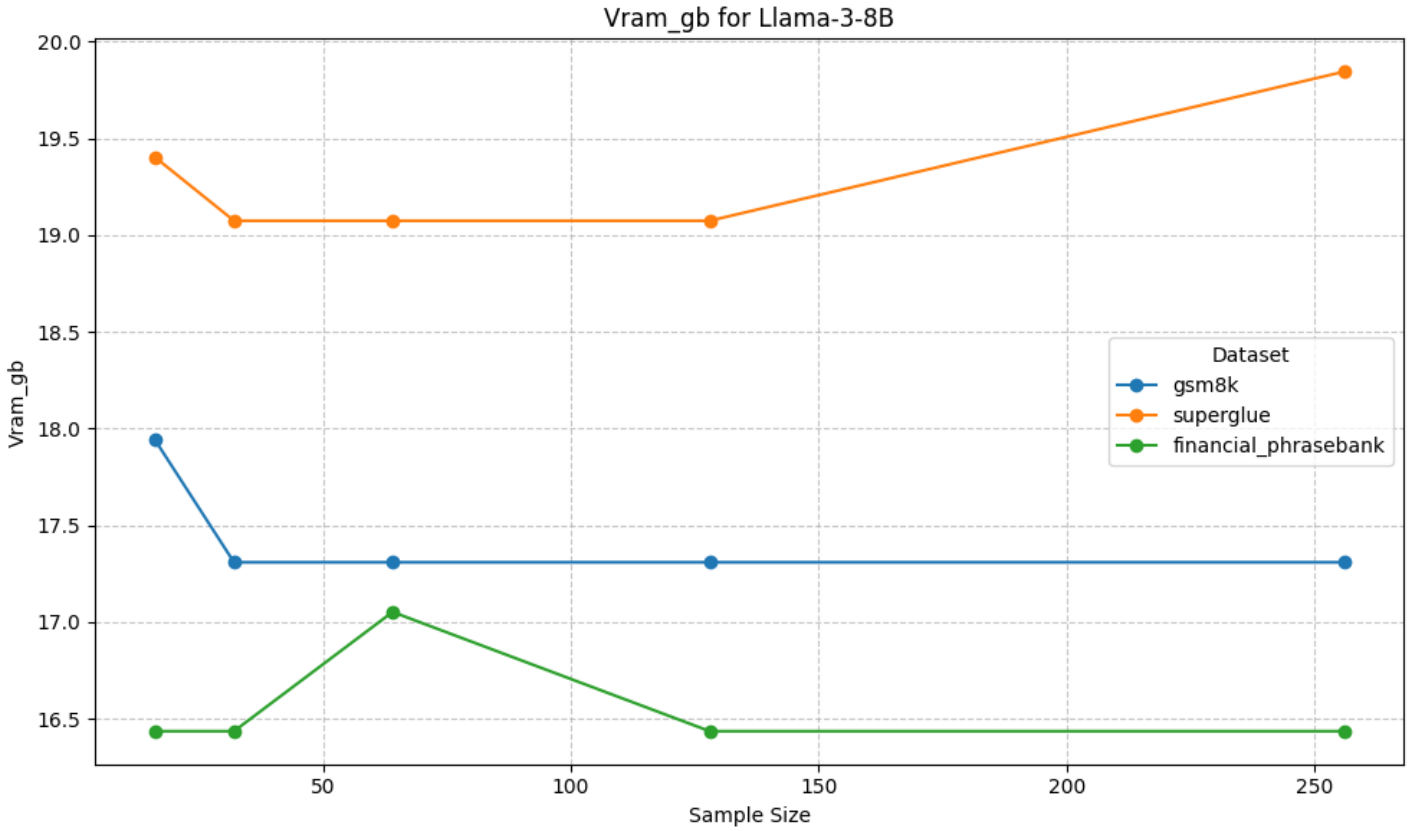

Latency for Llama-3-8B
Latency
Sample Size
3.0
2.5
2.0
1.5
1.0
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

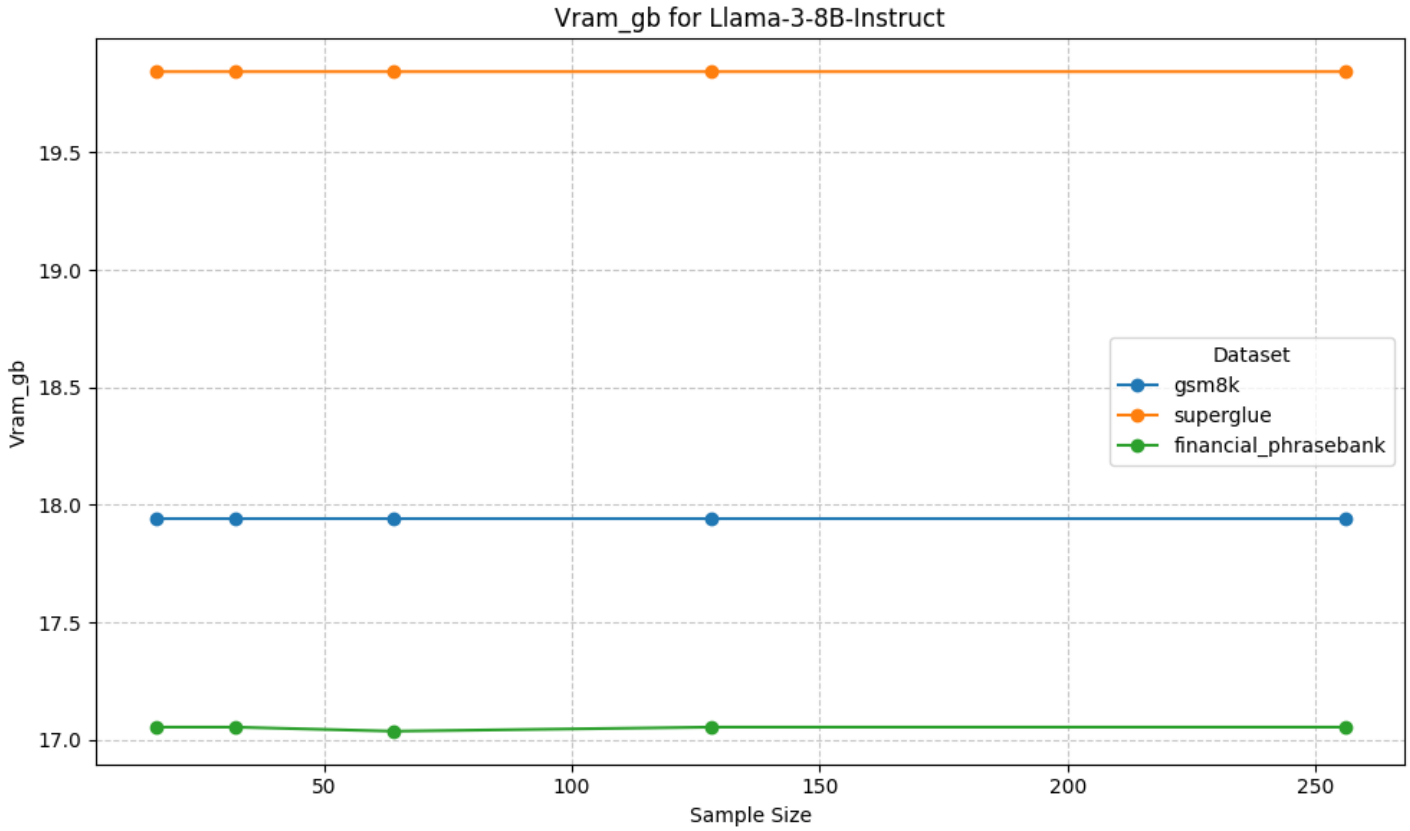

Latency for Llama-3-8B-Instruct
Latency
3.0
2.5
2.0
1.5
1.0
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank

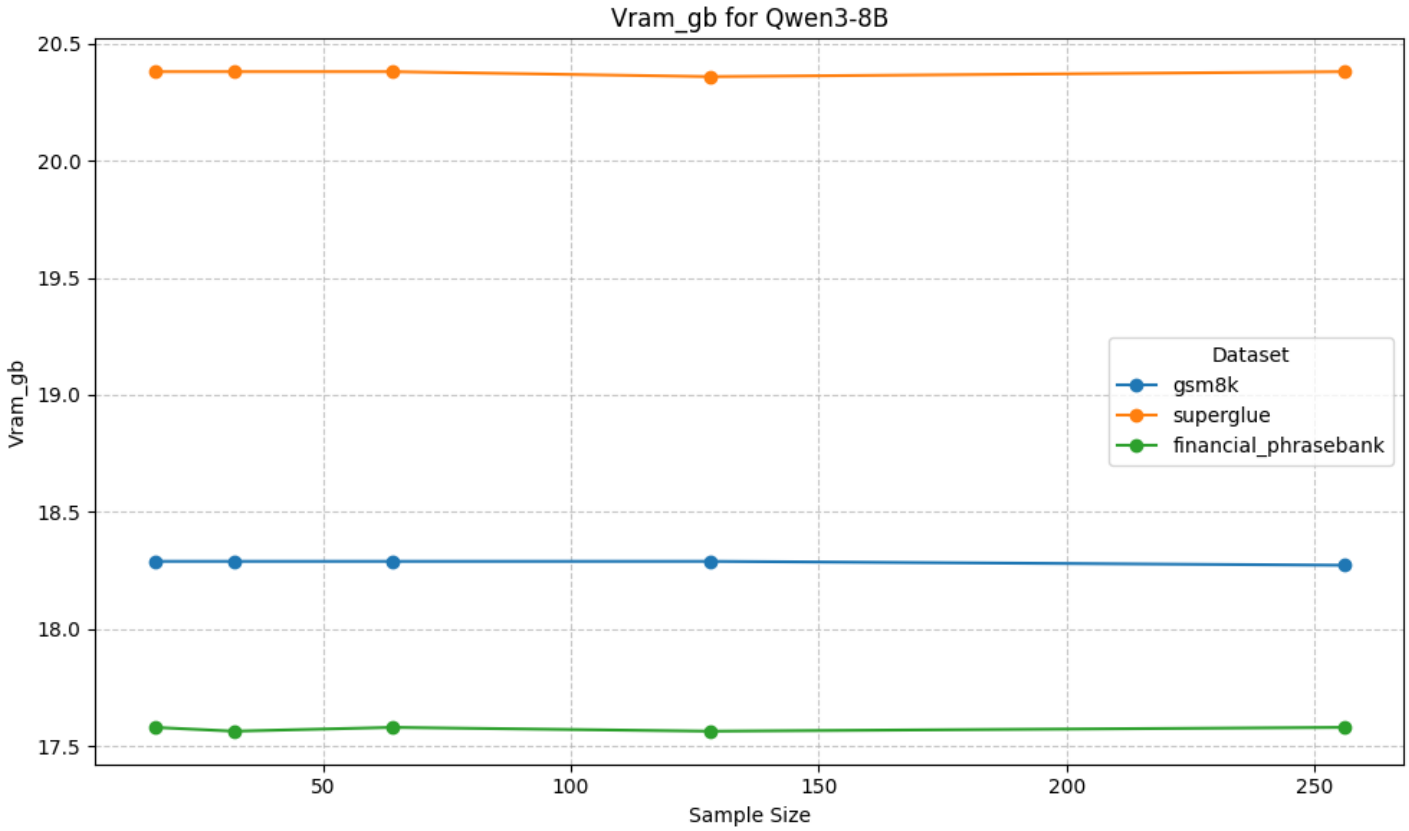

Latency for Qwen3-8B
Latency
Sample Size
4.0
3.5
3.0
2.5
2.0
1.5
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

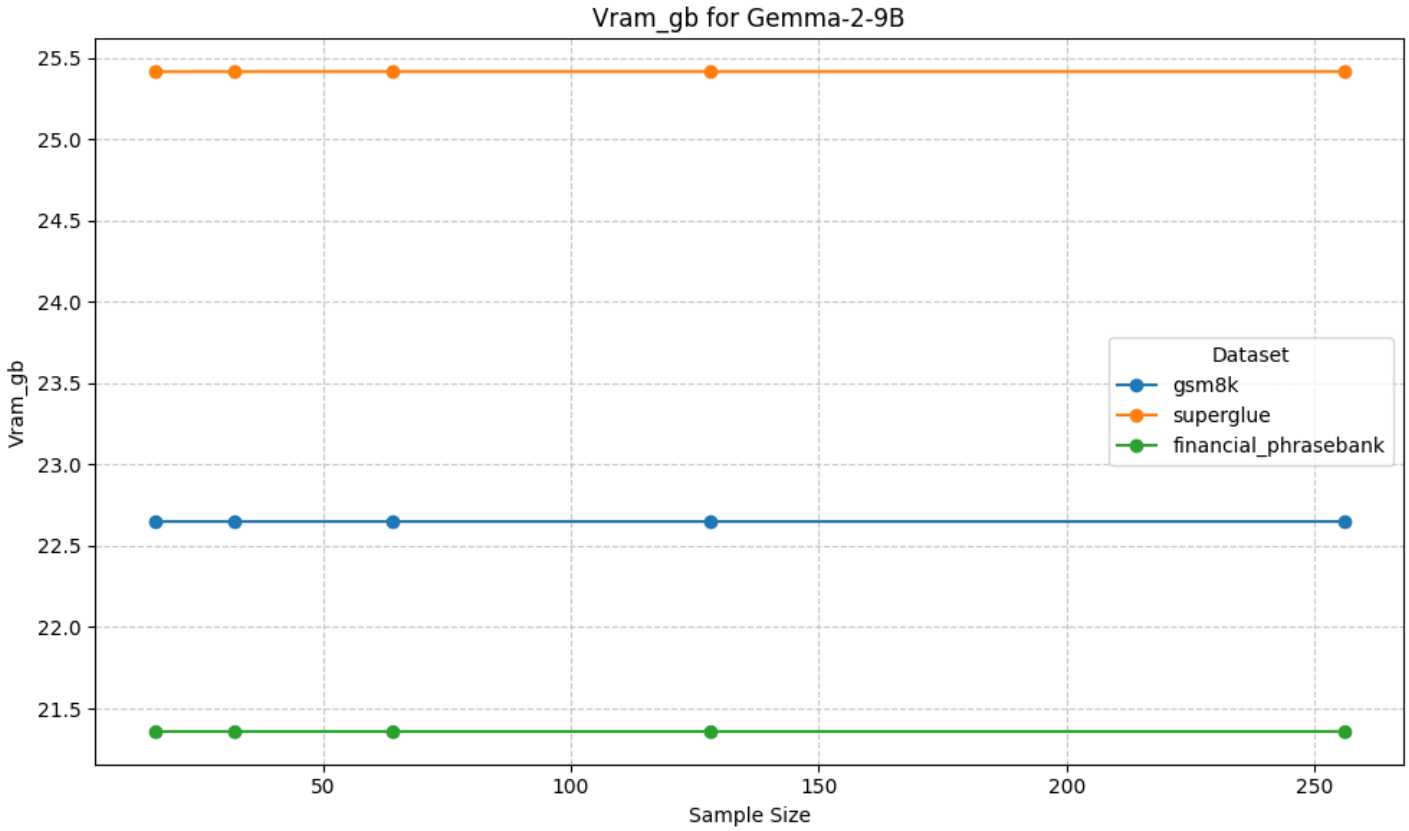

Latency for Gemma-2-9B
Latency
Sample Size
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

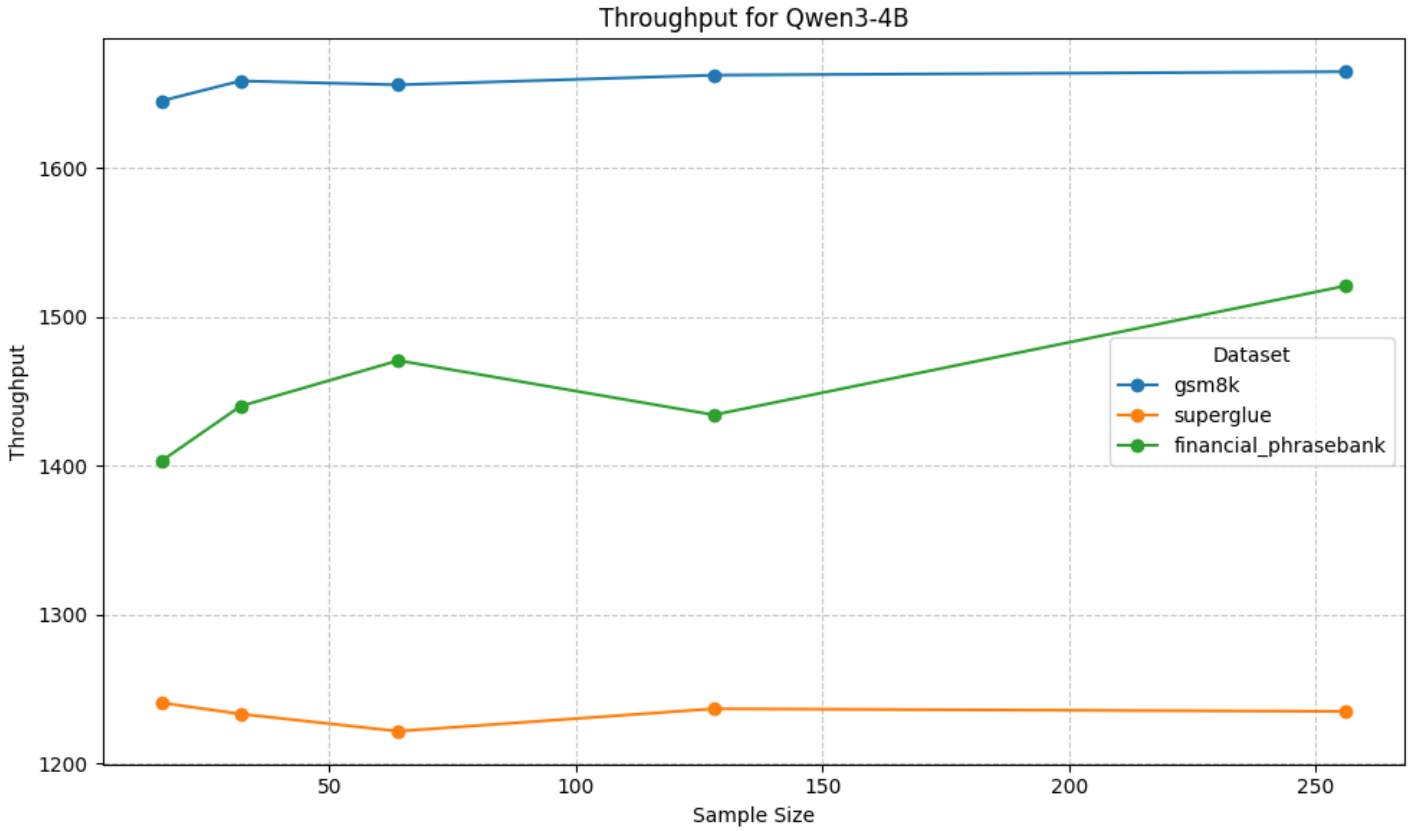

Vram_gb for Qwen3-4B
Vram_gb
Sample Size
12.5
12.0
11.5
11.0
10.5
10.0
9.5
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

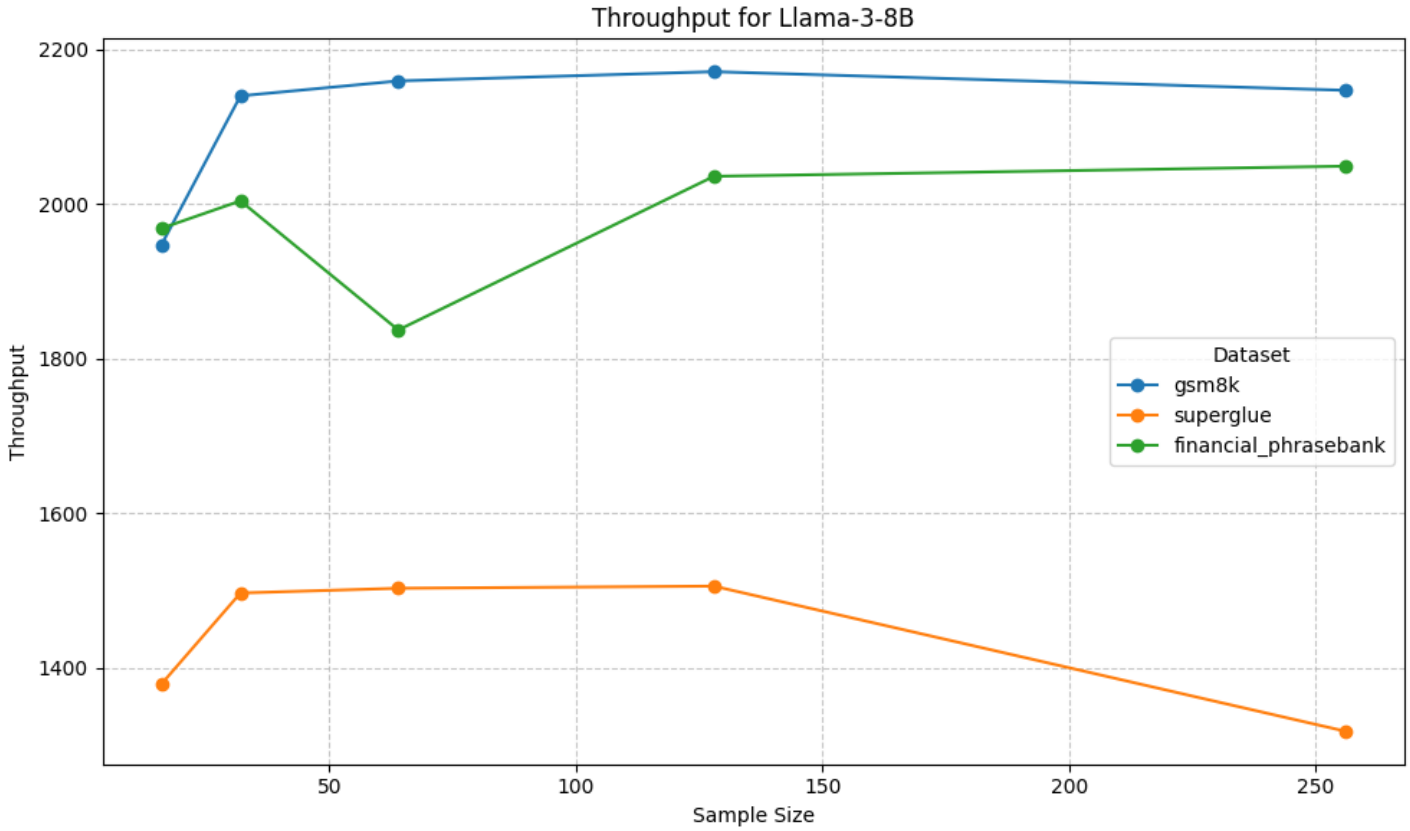

Vram_gb for Llama-3-8B
Vram_gb
Sample Size
20.0
19.5
19.0
18.5
18.0
17.5
17.0
16.5
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

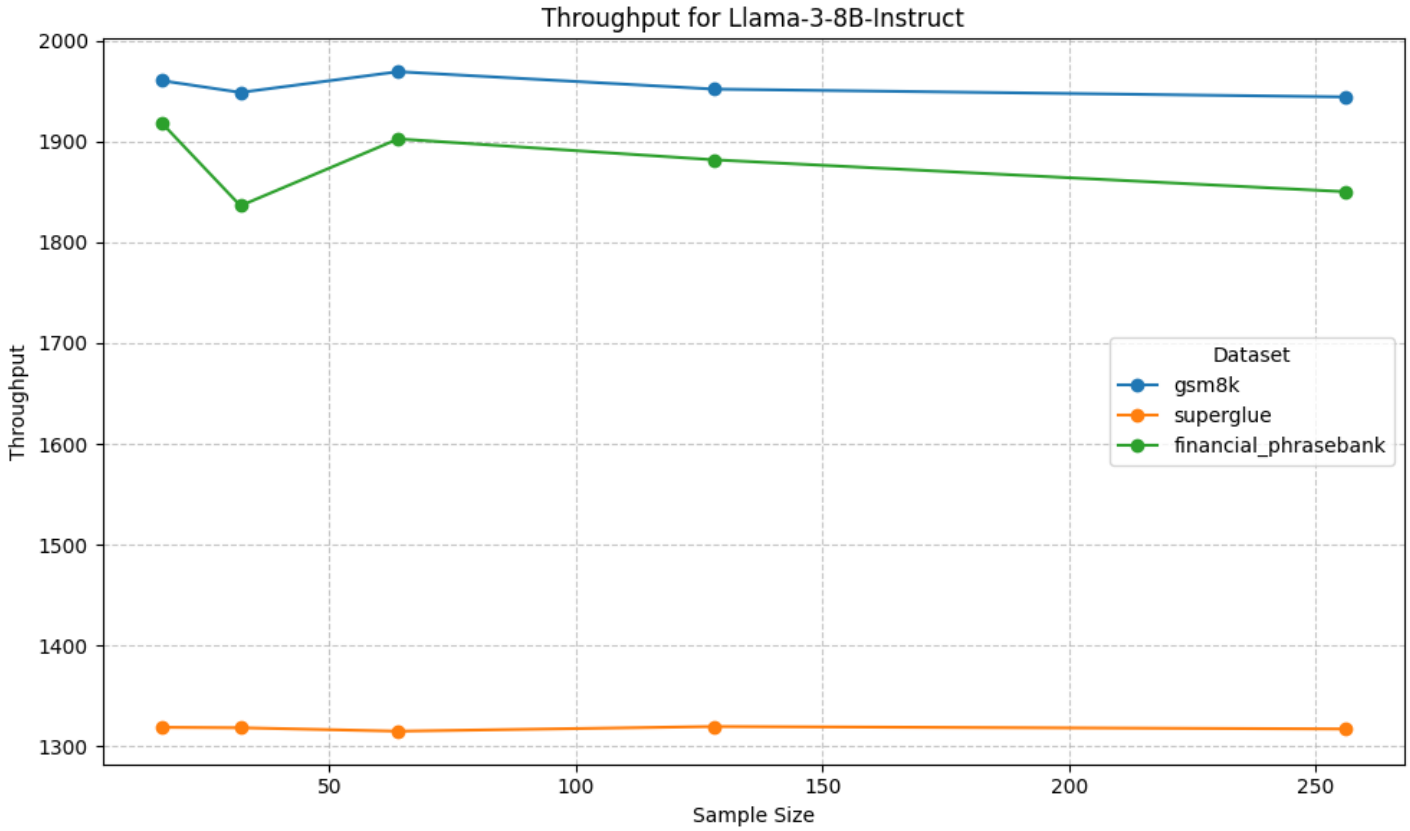

Vram_gb for Llama-3-8B-Instruct
Vram_gb
19.5
19.0
18.5
18.0
17.5
17.0
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank

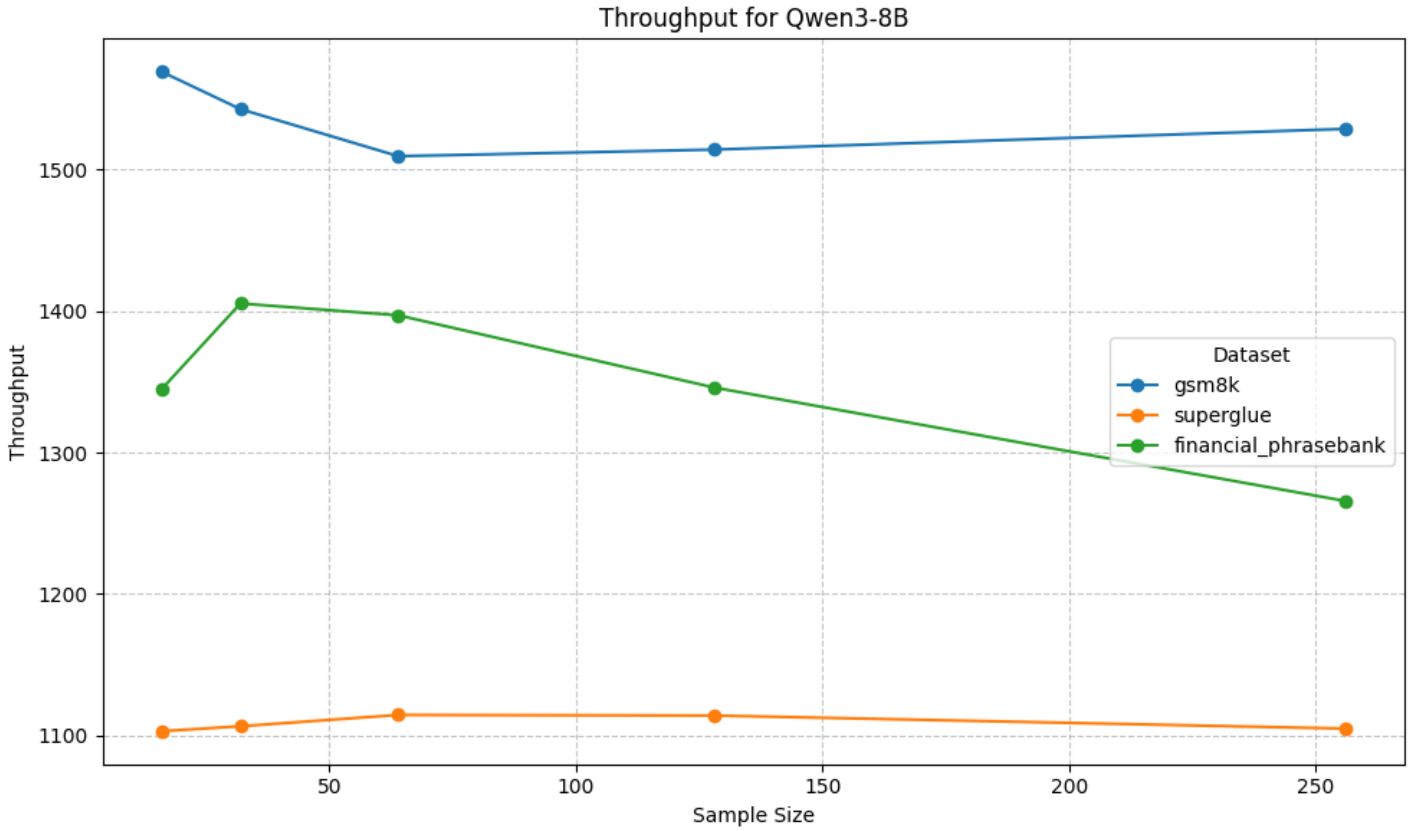

Vram_gb for Qwen3-8B
Vram_gb
Sample Size
20.5
20.0
19.5
19.0
18.5
18.0
17.5
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

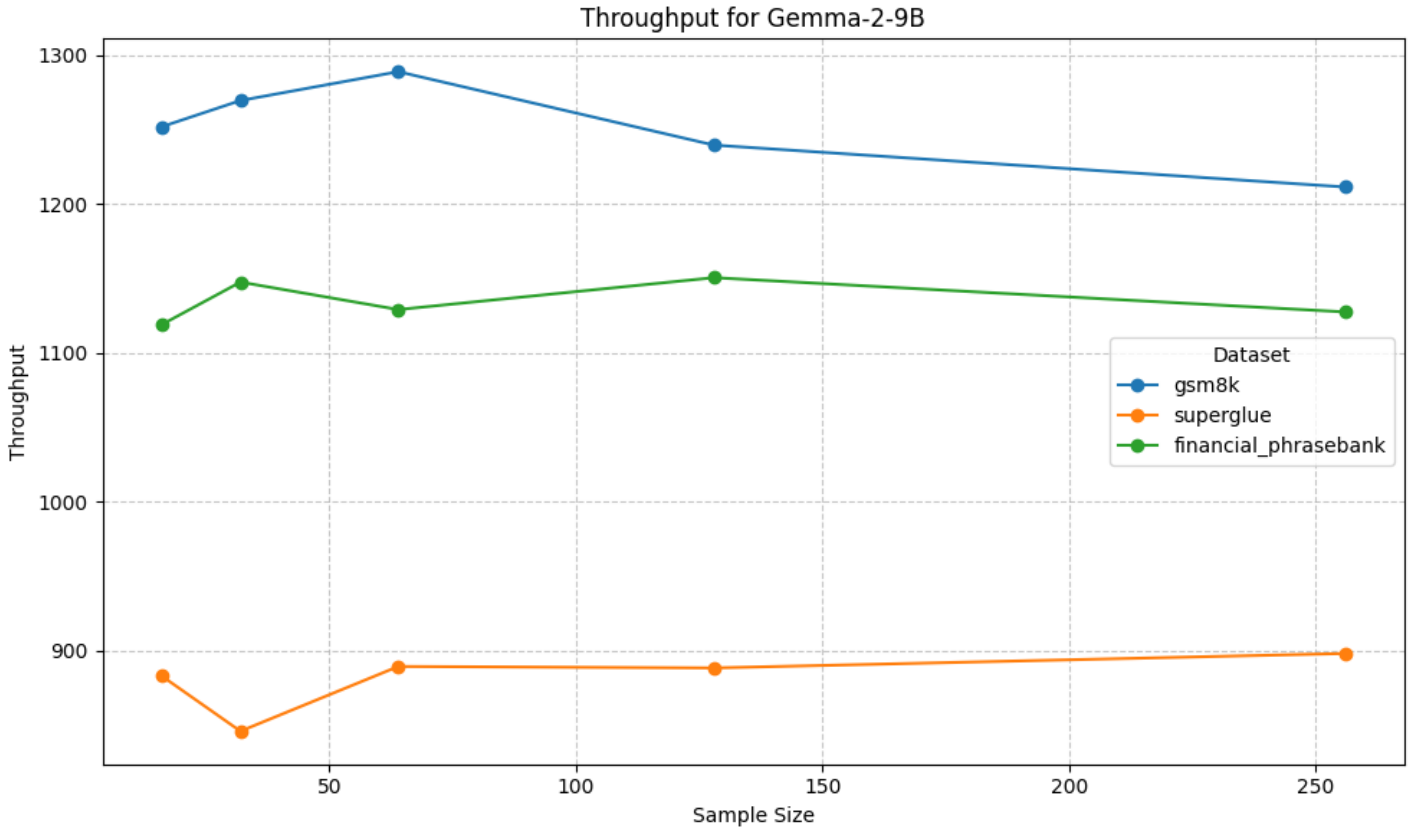

Vram_gb for Gemma-2-9B
Vram_gb
Sample Size
25.5
25.0
24.5
24.0
23.5
23.0
22.5
22.0
21.5
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

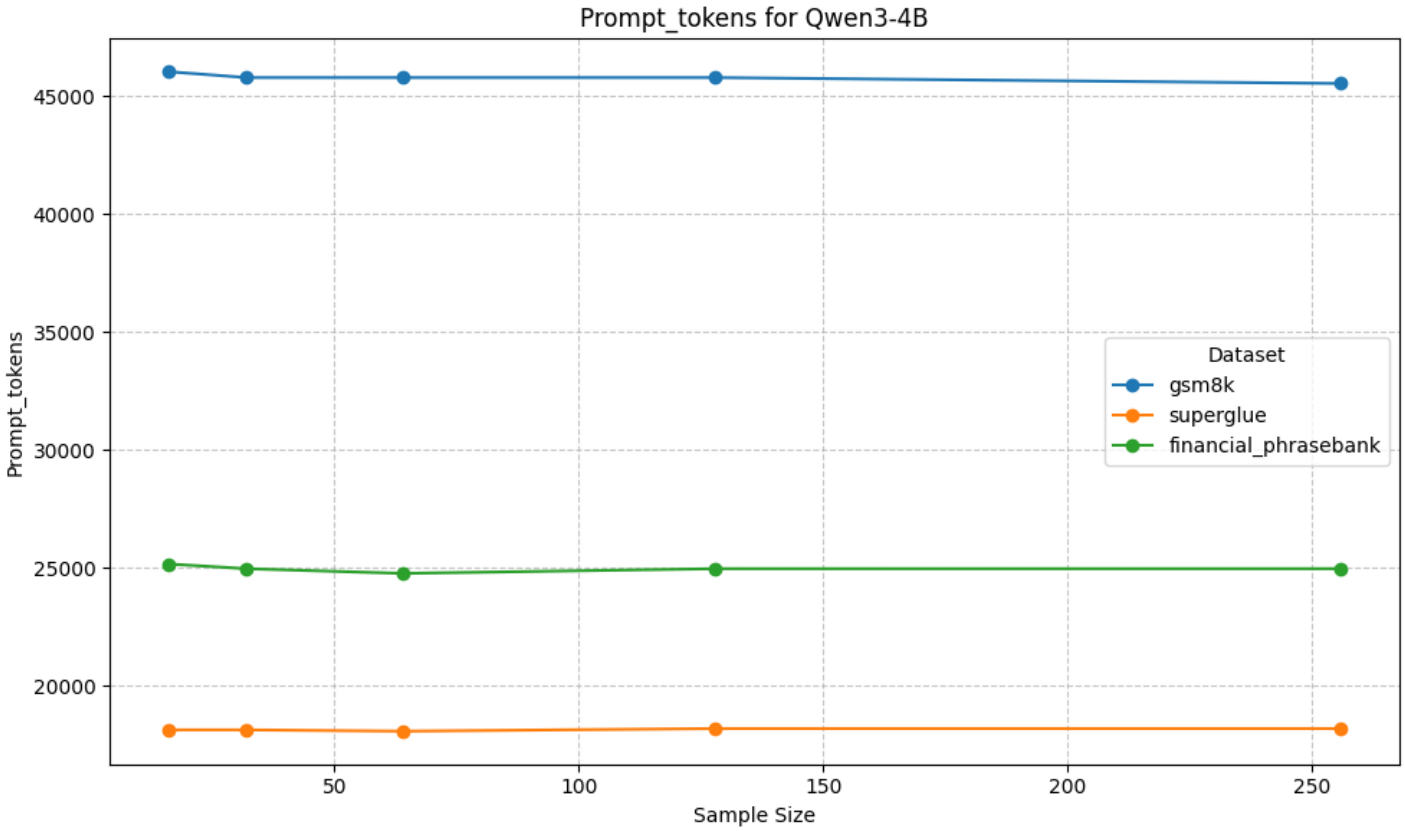

Throughput for Qwen3-4B
Throughput
Sample Size
1200
1300
1400
1500
1600
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

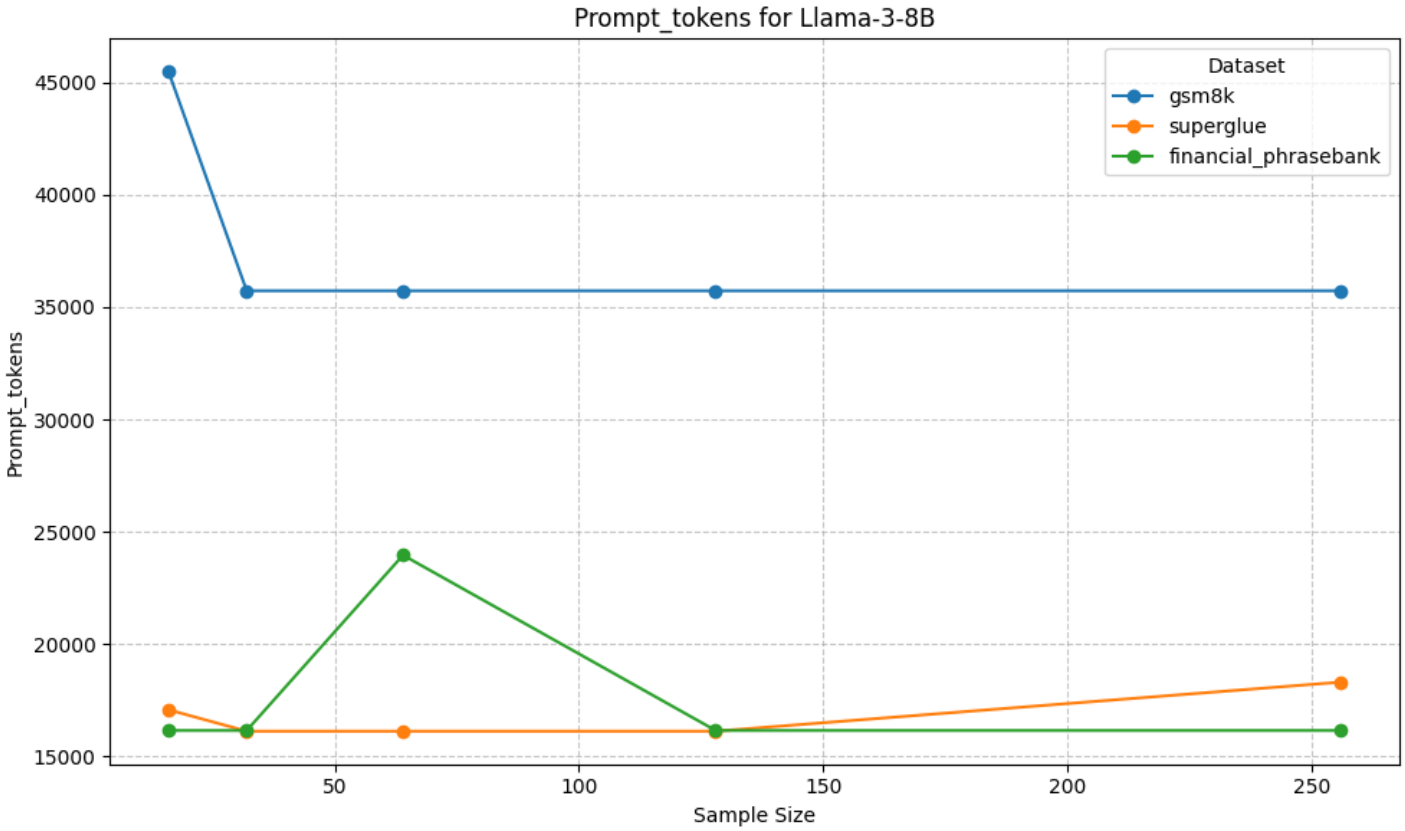

Throughput for Llama-3-8B
Throughput
Sample Size
2200
2000
1800
1600
1400
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

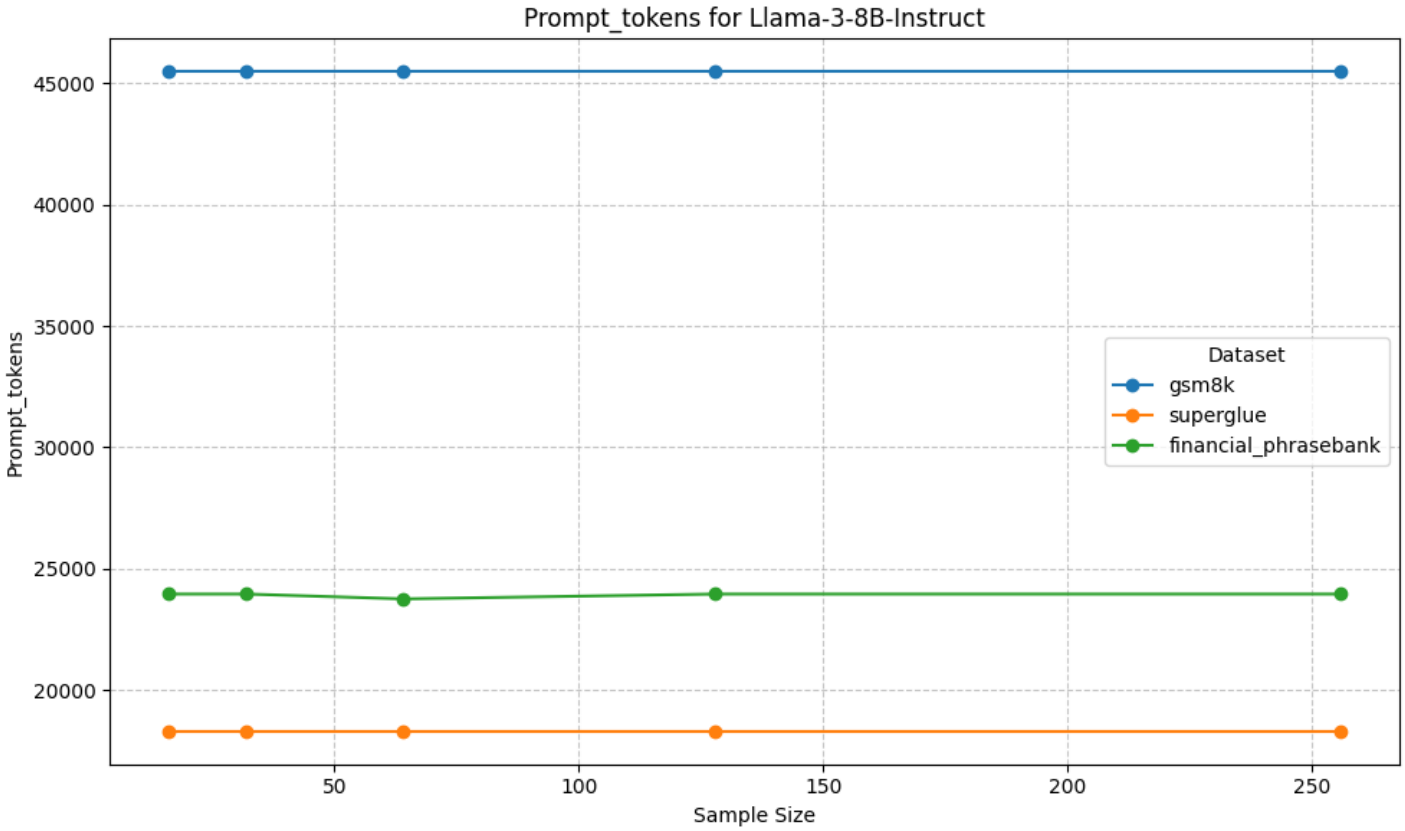

Throughput for Llama-3-8B-Instruct
Throughput
Sample Size
2000
1900
1800
1700
1600
1500
1400
1300
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank

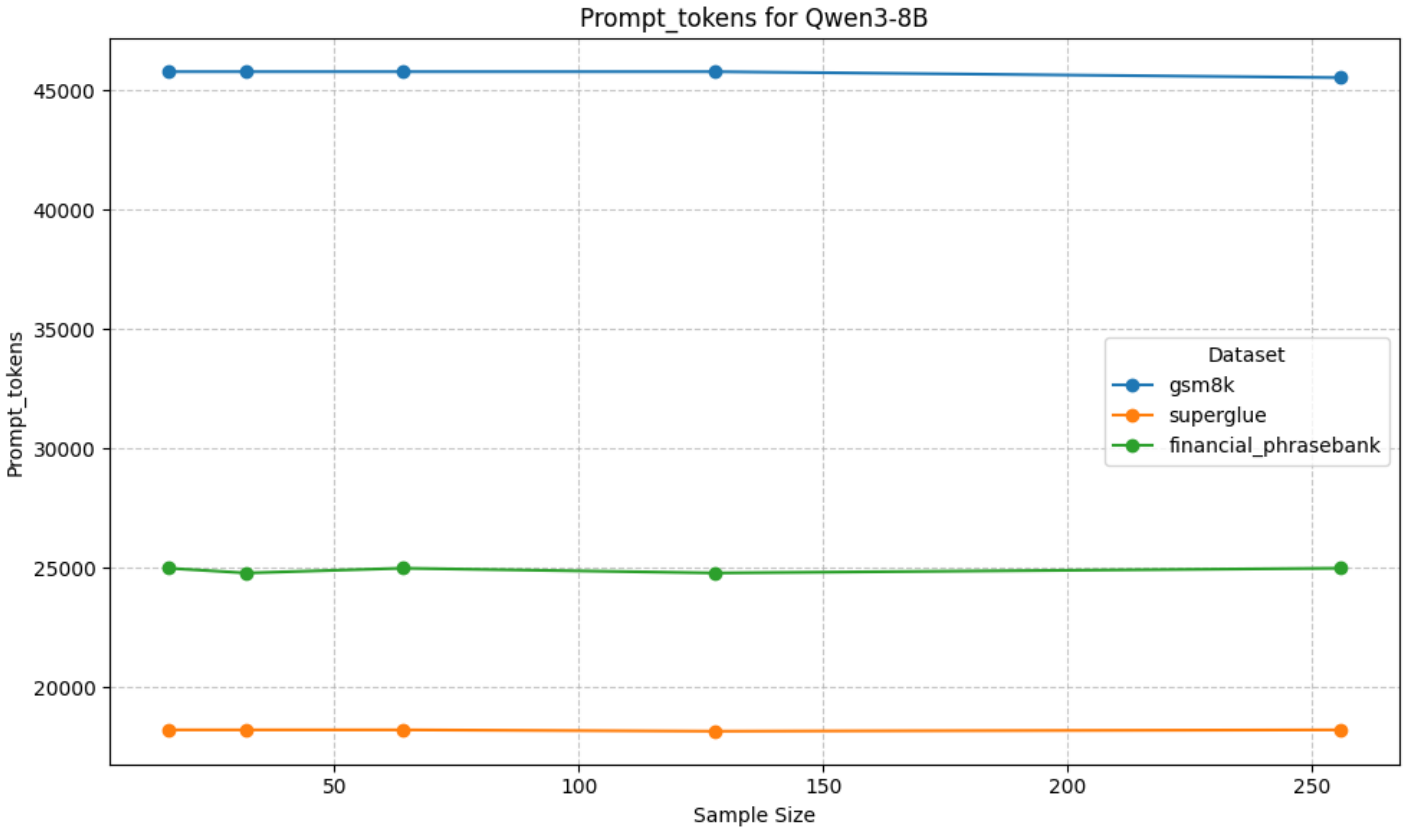

Throughput for Qwen3-8B
Throughput
1500
1400
1300
1200
1100
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank

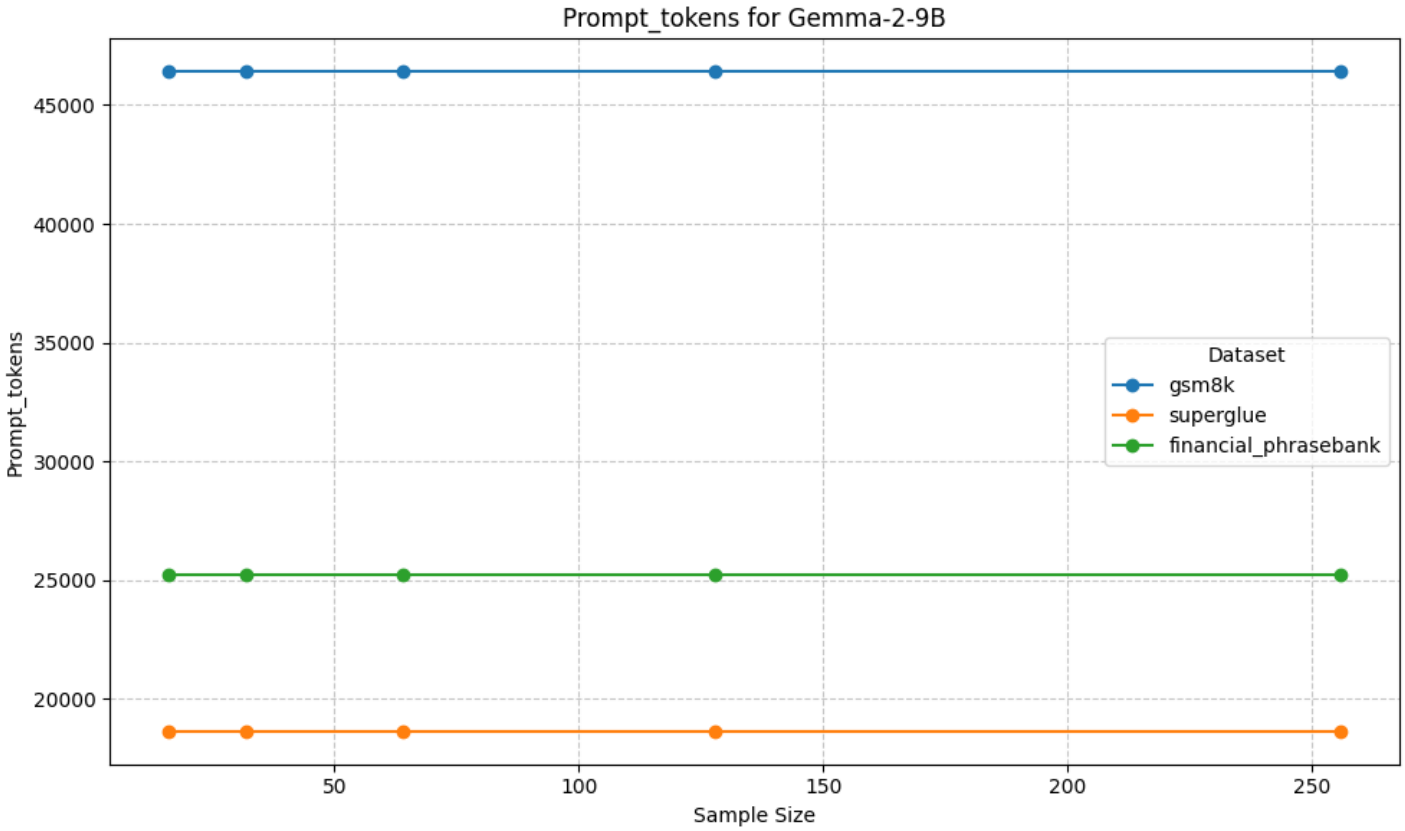

Throughput for Gemma-2-9B
Throughput
Sample Size
1300
1200
1100
1000
900
50
100
150
200
250
Dataset
gsm8k
superglue
financial_phrasebank


Prompt_tokens for Qwen3-4B
Prompt_tokens
45000
40000
35000
30000
25000
20000
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank


Prompt_tokens for Llama-3-8B
Dataset
gsm8k
superglue
financial_phrasebank
Prompt_tokens
45000
40000
35000
30000
25000
20000
15000
50
100
150
200
250
Sample Size


Prompt_tokens for Llama-3-8B-Instruct
Prompt_tokens
45000
40000
35000
30000
25000
20000
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank


Prompt_tokens for Qwen3-8B
Prompt_tokens
45000
40000
35000
30000
25000
20000
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank


Prompt_tokens for Gemma-2-9B
Prompt_tokens
45000
40000
35000
30000
25000
20000
50
100
150
200
250
Sample Size
Dataset
gsm8k
superglue
financial_phrasebank